Noakhali Science and Technology University
Department of Computer Science & Telecommunication Engineering
4th Year 1st Term B.Sc. (Engg.) Session: 2020-21 Final Examination-August, 2026
Course Code: CSTE- 4101 Course Title: Computer Graphics and Animations
Time: 4 hours. Total Marks: 70

[Answer any seven of the following questions. Figures in the right-hand margin indicate full marks]

1 a) Define Computer Graphics. What are the core components of a modern interactive graphics pipeline? 3

b) Compare raster graphics and vector graphics in terms of data representation, memory requirements, and suitable applications. Provide one example of a file format for each type. 3

c) An image has a resolution of 2560 x 1440 pixels. If the system uses a 24-bit RGB color model, what is the total size of the raw frame buffer in megabytes (MB)? 2

d) A system first processes a medical X-ray to enhance contrast and then uses a computer-generated overlay to mark a tumor location. Identify the image processing and computer graphics tasks in this scenario and justify your answer. 2

2 a) Define scan conversion. Describe the role of the decision parameter in the Bresenham line drawing algorithm. 3

b) Compare the DDA and Bresenham line drawing algorithms in terms of the operations used (floating-point vs. integer), accuracy, and performance. 3

c) Use the Bresenham line drawing algorithm to determine the pixels for a line from *(1,1)* to *(6,4)*. Calculate the values of *dx*, *dy*, the decision parameters at each step, and list the plotted pixels. 4

3 a) Explain the concept of 8-way symmetry in the midpoint circle drawing algorithm. Write the eight symmetric points generated from an initial point *(x,y)*. 3

b) How does the midpoint circle algorithm choose between the East (E) and Southeast (SE) candidate pixels for the next point on the circle? Explain the role of the decision parameter. 3

c) For a circle of radius $r=5$ centered at the origin, use $p0=1-r$. Starting at $(0,r)$, calculate the initial decision parameter $p0$ and determine the first four decisions (E or SE). 4

4 a) What is the rotation about the origin in 2D? Find the matrix that represents rotation of an object by 30^0 about the origin. 1.5 +2

b) Magnify the triangle with vertices A (0, 0), B (1, 1), and C (5, 2) to twice its size while keeping C (5, 2) fixed. 4

c) Describe the matrix representation of the basic 2D shearing transformations. 2.5

5 a) Using a column-vector convention, transform the point $P=(3,2)$ by first rotating it 90° counterclockwise about the origin and then scaling it by $sx=2$ and $sy=1$. Find the composite transformation matrix and calculate the final point P'. 4

b) What is the purpose of window-to-viewport mapping in the 2D viewing pipeline? Briefly explain the concept of clipping and why it is performed before mapping. 3

c) A world window is defined as $[50,150]\times[25,75]$. It is mapped to a viewport of $[100,500]\times[50,250]$. Calculate the viewport coordinates of the point $P=(100,50)$. 3

6 a) Explain the Cohen–Sutherland line-clipping algorithm using region (out-)codes, with a suitable diagram. ~~3~~4

b) Using the Cohen-Sutherland clipping algorithm, clip the line joining (5,20) and (60,40) against the clipping window bounded by $x_{min}=10$, $y_{min}=10$, $x_{max}=50$, $y_{max}=50$.. ~~7~~6

7 a) Describe rotation and reflection in 3D transformation. What is the difference between Cavalier and Cabinet Projection? 5

b) Find the transformation for Cavalier projection with $\theta = 45°$. Draw the projection of the unit cube for the transformation. 5

8 a) Why is a 4×4 matrix necessary for representing 3D affine transformations? Represent a 3D point (x,y,z) and a 3D vector (i,j,k) in homogeneous coordinates. 3

b) Distinguish between local coordinates, world coordinates, and transformed coordinates. Use the example of placing a chair model in a room scene. 3

c) Rotate the point $P=(1,2,3)$ by 90 degrees clockwise (i.e., $-90°$) about the Y-axis. Calculate the new point P'. 4

9 a) Define a perspective frustum. Why must clipping be performed before the perspective division and viewport mapping steps? 2

b) What is the purpose of hidden-surface removal in 3D rendering? Explain how the Z-buffer algorithm determines which surface is visible at each pixel. 3

c) Explain the primary problem of the Painter's Algorithm. Describe a situation where it fails and explain why the Z-buffer algorithm is generally more robust. 2

d) Evaluate the cubic Bézier curve at $t=0.5$ for the control points $P_0=(0,0)$, $P_1=(1,4)$, $P_2=(3,4)$, and $P_3=(4,0)$. You may use either the Bernstein polynomial or the de Casteljau algorithm. 3